

RESEARCH INTERESTS

- Distributional data analysis and optimal transport, with applications to wearable device data (current active research)
- Dimension reduction of compositional data, with applications to microbiome data
- Non-Euclidean data analysis with geometric constraints

PROFESSIONAL POSITIONS

Postdoctoral Research Fellow

09/2024 –

Department of Biostatistics, University of Michigan, MI, USA
Supervisor: Irina Gaynanova

BK21 Postdoctoral Research Fellow

03/2024 – 08/2024

Natural Science Research Institute, KAIST, Daejeon, Korea
Supervisor: Cheolwoo Park

EDUCATION

Ph.D. in Mathematical Sciences

03/2018 – 02/2024

KAIST, Daejeon, Korea

Thesis: “Kernel Methods for Compositional Data and Dimensionality Reduction”

(Co)Advisors: Cheolwoo Park, Jeongyoun Ahn

B.S. in Mathematics

03/2013 – 02/2018

Korea University, Seoul, Korea

EXPERIENCE

Research Assistant in Statistical Learning

KAIST

Under the supervision of Prof. Cheolwoo Park and Jeongyoun Ahn

08/2021 – 02/2024

– Kernel methods, compositional data, dimension reduction

Technical Research Personnel

KAIST

Military duty in South Korea

03/2020 – 02/2023

Research Assistant in Algebraic Geometry

KAIST

Under the supervision of Prof. Sijong Kwak

09/2018 – 07/2021

– Projective algebraic geometry, syzygies, applied algebraic geometry

University Financial Engineering Association (U.FE.A)

Seoul, Korea

Team leader (Master's-level financial engineering & mathematics studies)

03/2016 – 08/2017

AWARDS AND SCHOLARSHIPS

Competitive Travel/Paper Awards

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|--|------|
| • IMS New Researchers Travel Award | 2026 |
| • 1st IMS International New Researchers Conference, Seville, Spain | 2025 |
| • Best Ph.D. Dissertation Award, Korean Statistical Society, 2 awardees per year | 2025 |

Other Awards/Scholarships

- **Presentation Award for Graduate Students**, Korean Statistical Society; prize: 300,000 (KRW) 2022
- **The Outstanding Teaching Assistant Award**, Calculus II, KAIST Fall, 2020
- **University Students Contest of Mathematics**, Silver Awards, Korean Mathematical Society (KMS) 2016, 2017
- **Presidential Science Undergraduate Fellowship**, fully funded for 8 semesters 2013–2018
- **The Korean Mathematical Olympiad (KMO)** 2nd round of the middle school division, Gold Awards 2009

RESEARCH PAPERS

Preprints/Submitted:

9. **Park, J.** and Gaynanova, I. (2026+) “Fréchet regression of multivariate distributions with nonparanormal transport.” Submitted. Preprint available on [arXiv:2603.07014](https://arxiv.org/abs/2603.07014).
 - 2026 IMS New Researchers Travel Award
8. Shao, E., **Park, J.**, Punjabi, N., Jiang, H, and Gaynanova, I. (2026+) “Fast distance computation between multivariate distributions via nonparanormal Transport.” Preprint available on [arXiv:2603.00322](https://arxiv.org/abs/2603.00322).
7. Xu, C., Schmidt, B.M., Krambrink, A., **Park, J.**, Song, P.X.-K., Jones, T.L.Z., Holmes, C., Chen, K., Ye, W., Kolenic, G., Roy, S., Sen, C., Pop-Busui, R., Spino, C., and Gaynanova, I. (2026+) “Profiling Longitudinal Wound Healing: Analysis of Healing Rates and the Predictive Power of Wound Area and Duration in the Diabetic Foot Consortium.” Submitted.
6. **Park, J.**, Park, C., and Ahn, J. (2026+) “A geometry-preserving approach to interpretable dimension reduction of compositional data.” Submitted. Preprint available on [arXiv:2509.05563](https://arxiv.org/abs/2509.05563).
5. **Park, J.**, Kok, N., and Gaynanova, I. (2026+) “Beyond fixed thresholds: optimizing summaries of wearable device data via piecewise linearization of quantile functions.” Submitted. Preprint available on [arXiv:2501.11777](https://arxiv.org/abs/2501.11777).

Peer-reviewed publications:

4. Yeon, K., Gao, F., Kim, R., Lee, D., **Park, J.**, Brown, D.A., and Park, C. (2026+) “A Bayesian Approach to Semi-Supervised Domain Adaptation in Streaming Data.” *Journal of Statistical Computation and Simulation*, Accepted.
3. **Park, J.**, Ahn, J., and Park, C. (2023), “Kernel Sufficient Dimension Reduction and Variable Selection for Compositional Data via Amalgamation.” *International Conference on Machine Learning (ICML)*, pp. 27034-27047, PMLR
Link: <https://proceedings.mlr.press/v202/park23a.html>.
2. Kang, I., Choi, H., Yoon, Y.-J., **Park, J.**, Kwon, S.-S., and Park, C. (2023), “Frechet Distance-Based Cluster Analysis for Multi-Dimensional Functional Data.” *Statistics and Computing*, 33(4), 75.
Link: <https://doi.org/10.1007/s11222-023-10237-z>
1. **Park, J.**, Yoon, C., Park, C., and Ahn, J. (2022), “Kernel Methods for Radial Transformed Compositional Data with Many Zeros.” *International Conference on Machine Learning (ICML)*, pp. 17458-17472, PMLR
Link: <https://proceedings.mlr.press/v162/park22d.html>.

TALKS

Invited talks:

- (Scheduled) **IMSI Long Program**, 2 weeks invited visit to Connectomics program Dec, 2026
- (Scheduled) International Conference on Computational Statistics (CompStat) 08/2026
- **1st IMS International New Researchers Conference** 12/2025
 - Title: Fréchet regression of multivariate distributions under the semiparametric nonparanormal model
- **KAIST, Graduate School of Data Science** 07/2025
 - Title: Distributional data approaches for wearable device data: optimal thresholds and beyond

Contributed talks/posters:

- **Mobile Technologies Research Innovation Collaborative (MeTRIC) Symposium** (poster) 01/2026
 - Title: Beyond fixed thresholds: optimizing summaries of wearable device data
- **IMS International Conference on Statistics and Data Science (ICSIDS)** 12/2025
 - Title: Fast Fréchet regression of multivariate distributions with application to wearable device data
- **Joint Statistical Meetings** 08/2025
 - Title: Beyond fixed thresholds: optimizing summaries of wearable device data
- **Joint Statistical Meetings** 08/2024
 - Title: Interpretable dimension reduction for compositional data
- **Winter Conference, the Korean Statistical Society** 12/2023
 - Title: Interpretable composition-to-composition dimension reduction via conditional covariance operator
- **International Conference on Machine Learning (ICML)** (poster) 07/2023
 - Title: Kernel sufficient dimension reduction and variable selection for compositional data via Amalgamation
- **Summer Conference, the Korean Statistical Society** 06/2023
 - Title: Kernel sufficient dimension reduction and variable selection for compositional data via Amalgamation
- **KAIST Mathematical Sciences Graduate student Seminar** 10/2022
 - Title: Kernel methods for radial transformed compositional data with many zeros
- **International Conference on Machine Learning (ICML)** (spotlight talk) 06/2022
 - Title: Kernel methods for radial transformed compositional data with many zeros
 - Presented also at 2022 Summer Conference, the Korean Statistical Society, (**presentation award**)

TEACHING

- **Teaching Assistant** at KAIST (selected list)
 - Statistical Data Science Practice (DS516) Spring 2023
 - Probability and Statistics (MAS250) Fall 2021
 - Abstract Algebra I (MAS311) Spring 2021
 - Mathematical Statistics (MAS355) Fall 2019
 - Matrix Group Theory (MAS435) Spring 2019
 - Gave a guest lecture on connectedness of Lie groups (in English)
 - Abstract Algebra II (MAS312) Fall 2018
 - Gave several guest lectures throughout the semester

ACADEMIC SERVICES

- **Journal Refereeing**
 - Journal of Royal Statistical Society, Series B (1); Annals of Applied Statistics (2); Biometrics (3); WIREs Computational Statistics (1)

COMPUTING SKILLS

- Python (for machine learning, statistics), R (for statistics), previous experiences with Matlab and C
- Optimization acceleration: with deep learning frameworks (**TensorFlow** and **Pytorch**), with just-in-time compiler (**numba**)
- Linux environment and high-performance computing (HPC) cluster systems

LANGUAGES

- English
- Korean

HOBBIES

- Singing, better with playing guitar.
- Running, hiking, and climbing.